

From the Curtain to Kansas: Conducting Wizard-of-Oz Studies in the Wild

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ABSTRACT

Conversational agents have been touted for their potential to support individuals over time as health coaches or personal assistants, but have yet to live up to this potential. Wizard-of-oz (WOz) methods enable researchers to test early prototypes of conversational applications before they are fully implemented, with a human “wizard” filling in the gaps in functionality. Current WOz methods, however, are more commonly used for studies in a lab setting, rather than deployment studies, which more accurately capture users’ interactions in-the-wild. We argue for the need for WOz methods for deployment studies that address key challenges, namely the need for easy-to-prototype technology that works reliably in the wild, as well as the user’s expectations for 24/7 availability. We describe an initial approach that begins to address these challenges, as well as the insights gleaned from a two-week WOz study of *t2.coach*, a conversational agent health coach for diabetes self-management. We argue that the findings from our WOz study could not have been identified from a lab usability study with the same prototype, and the need for the research community to further develop methods for WOz deployment studies.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods.

KEYWORDS

wizard-of-oz, conversational agents, chatbots, deployment study

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1 INTRODUCTION

Interest in conversational agents has been increasing in recent years, both in the research community and in industry. The potential of conversational agents — sometimes referred to as chatbots — has been touted since Eliza, the first chatbot, was introduced in the 1960s [29]. Researchers have argued that conversational interaction is a natural mode of communication, and could be used to design

technology that imbues human elements into the interaction. This could be particularly useful for applications like health coaching, mental health counseling, or personal assistants, where the user and agent build rapport over time in a longitudinal relationship [1]. Unfortunately, many current conversational agents fall short of this imagined potential, and there is still a need for new systems that bridge this gap [9, 18, 24]. Furthermore, there is a need for design methods that can enable evaluation of new chatbots early in the design process without investing time and effort in development of fully functional systems.

One particularly useful technique for designing new conversational agents is wizard-of-oz (WOz) prototyping [5, 13]. With WOz, a researcher works behind the scenes to create the illusion of a fully functioning system, even if the system is only partially implemented. While WOz methods have been used for many types of interactive systems, they particularly lend themselves to conversational interaction: the human wizard can easily recognize and interpret users’ statements and requests and generate appropriate responses. WOz can enable prototyping and iterative refinement of a design before moving to implementation, which can be labor intensive, or in some cases require novel machine learning approaches or data sets.

However, the vast majority of WOz experiments reported thus far were carried out in a lab setting, rather than in deployment studies in-the-wild; yet there are many situations where a lab experiment may be inadequate to understand a user’s interactions with a system. For example, technology designed to help individuals adopt healthy behaviors needs to be studied over weeks, months, or years, because behavior change is a slow process that unfolds gradually overtime [2, 16, 22] and patterns of interaction may change as individuals gain more experience with the system [6]. Important design decisions about the length and form of conversational flows, the frequency of daily messaging, and even the tone and content of responses may yield different feedback in a lab setting than through continuous daily use. Setting the prototype in the wild, without researchers on hand to intervene and help, creates a more realistic testing scenario.

Adapting the WOz technique to a real-world setting, however, poses challenges, including the need for a lightweight system that works well outside the lab, an expectation of 24/7 availability of the agent, and the need to manage conversations with multiple users at once. Unfortunately, there are very few examples of WOz applied to longitudinal deployments or field studies in the HCI literature for researchers to leverage and build upon.

In this case study, we describe our experience conducting a WOz study of a conversational agent health coach, called *t2.coach*, designed to support individuals with type 2 diabetes (T2D) in selecting and achieving nutrition and physical activity goals through text-message exchanges. The WOz study with 13 individuals over two

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weeks was part of the iterative design of t2.coach, and we sought to understand how individuals with T2D and limited prior experience with health apps would engage with a chatbot coach. In particular, we had questions about how users would engage in longitudinal goal setting, how often users would engage with messaging, how many messages they would respond to in a single exchange, as well as the content and flow of interactions, which were based on a protocol for in-person goals setting (Brief Action Planning [10]). We found that substantial delays in responses from participants informed the need to considerably shorten the length of dialogs for goal setting in t2.coach. We use the findings from this study to suggest that the current literature in WOz methods is lacking in specifying an approach for longitudinal WOz studies, and that such methods can be highly informative in the design process for conversational agents.

2 RELATED WORK

The Wizard-of-Oz (WOz) approach originated, at least in name, from Kelley’s work on the OZ Paradigm for testing prototype conversational agents [7, 13]. With WOz, a user interacts with a conversational agent or other system, that appears to the user to be a fully functional prototype. The system, however, is not fully functioning. Instead, a human operator — dubbed the “wizard” — is working behind the scenes to make the system respond appropriately to the user’s input. The intention of including a WOz prototyping step in the iterative design of a system is that it allows designers to assess users’ reactions and impressions without needing to fully implement the system, and then re-implement it based on user feedback [5].

WOz can be applied with mixed degrees of involvement from the wizard [7]. In some cases, WOz can be an early step in the design process, to test design intuitions before anything at all has been developed. In other situations, the system can be almost fully functioning, and the wizard fills in one or two key gaps that would be particularly resource intensive to implement. Wherever it is applied in the design process, even though the wizard is a human-in-the-loop, the wizard is usually not meant to act as a human conversational partner, but rather follow a protocol or guidelines to respond in a manner that closely aligns with what could reasonably be implemented [19].

Because the human is emulating the behavior of an automated system, one potentially contentious question with WOz designs is the degree of deception employed: to what extent should researchers be upfront with the user that they are interacting with a human conversant, or practice a degree of deception to maintain the illusion of a fully functioning prototype. Bradley and colleagues presented evidence that the degree of deception may not impact users’ perceptions of the system [4], which suggests that deception may not be necessary to the validity of results from a WOz study.

Much of the seminal WOz research involved conversational agents, but it has since been applied to many other application areas like location-aware computing [17] and augmented reality [8]. In all of these application areas, most studies are conducted in a lab setting, not with a field or deployment study. Even those that do include a real-world research setting do so for a limited period of time. For example Robinson and colleagues described the

deployment of a smart speaker simulated with WOz in a public slum setting in India, but the deployment was short term and lasted hours, not days or weeks [27]. Focusing on brief study periods makes sense given how resource intensive WOz approaches can be — to simulate a chatbot, for instance, a human “wizard” needs to be available at all times to create the illusion of a functioning system. Indeed, a cadre of wizards available around-the-clock would defeat the intended purpose of WOz as a lightweight, low-cost prototyping technique. In light of this, there is a need for WOz methods for deployment studies that maintain the iterative prototype spirit of WOz in the lab.

It is also important to consider the durability and extensibility of WOz methods. Some researchers have introduced and evaluated complete systems to implement WOz prototypes [14]. However, Pettersson and Wik noted that many WOz prototyping systems are short-lived, and only see a few years of continued use in a research setting [23]. Many studies that use WOz methods offer scant details on their implementation approach, which makes it difficult for researchers to borrow from and build on prior approaches [26]. Instead of full-fledged prototyping systems, there is a need for more clearly articulated WOz methods.

3 T2.COACH WIZARD-OF-OZ PROTOTYPE

3.1 Initial Design of t2.coach

Next, we briefly describe the conversational agent that was the subject of our WOz study, t2.coach. We designed t2.coach as an intervention to support individuals living with type 2 diabetes. In particular, we leveraged health coaching as an approach to promote self-management. In-person health coaching can be effective, but there are not enough coaching practitioners to reach the growing population in need of support. Conversational agents are a natural choice for health coaching interventions because coaching is intended to be a conversational exchange between two parties, the coach and the client, in a relationship that develops and strengthens over time [22, 28].

In the initial design of t2.coach, we followed an established protocol, Brief Action Planning (BAP; [10]) as the basis for the scripted dialog flows. BAP defines a set of steps for health practitioners to guide an individual towards choosing a health goal and making a specific plan to achieve it. These goals could be either based on general nutritional guidelines, or personalized using computational analysis of individuals’ self-monitoring data [21]. Consistent with BAP, t2.coach included two primary dialogs: 1) a longer, weekly exchange to set a health goal, as well as 2) a shorter, daily, follow-up exchange to check in on goal progress, shown in Figure 1

In addition to daily, system-initiated dialogs, meant to promote behavior change and account ability, t2.coach also promoted engagement by encouraging users to initiate interactions. Users could ask questions and send messages to t2.coach throughout the day, and the chatbot would respond with answers similar to the types of question-answering offered by modern, commercially available agents (e.g. Siri, Alexa) [20], as well as tailored responses to certain requests, like diabetes-friendly recipes.

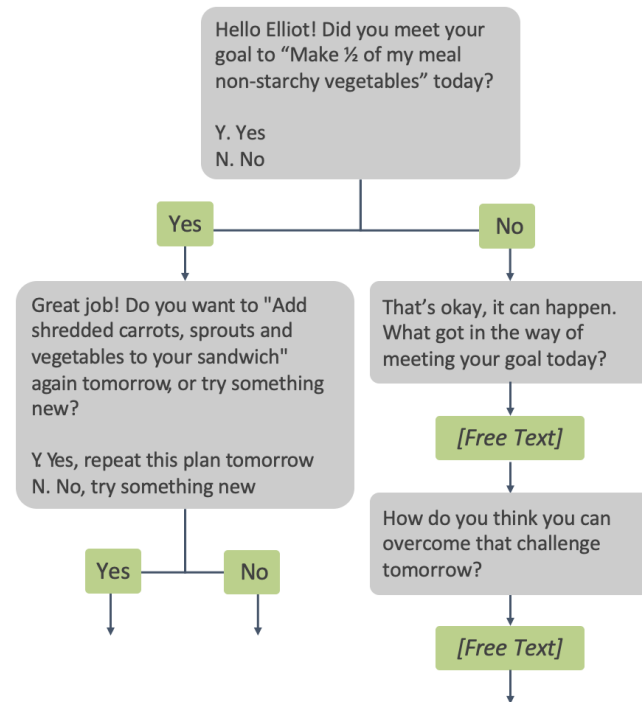


Figure 1: An example dialog tree from t2.coach for the daily check-in script

3.2 Approach to Wizard-of-oz in the Wild

We conducted a WOZ study with 13 participants with T2D recruited from Federally Qualified Health Centers in the New York City metro area. We were specifically interested in examining experiences of individuals within economically disadvantaged communities, who engage with self-management interventions differently from those in more affluent populations [25]. Participants were recruited in groups of 2-5, and the study ran for 2 weeks. The “wizard” strictly followed the same rules that the fully implemented chatbot would use respond, including the fallback response “I’m sorry, I didn’t understand” and re-prompting the question for off-script utterances. To simulate question-answering, the wizard ran requests through a classifier built with AWS Lex [30] to categorize queries to different types of requests including recipe requests or diabetes knowledge questions. General informational questions were searched verbatim in Google, with the additional keyword “diabetes,” and responses from Google snippets were sent with a link, after being approved by a CDE on the research team. If the question was unclassifiable, or the response was deemed inappropriate, the chatbot responded that it was unable to answer the question.

In designing and implementing the apparatus for WOZ messaging we took advantage of existing available technologies to the extent possible. t2.coach was designed as an SMS-based text messaging chatbot because of the ubiquity of text-message use in the population, and because of potential privacy and security issues with speech-to-text software [15]. For simplicity, we allowed the

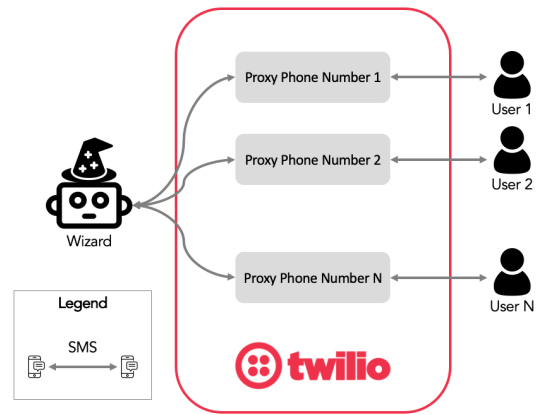


Figure 2: An illustration of the wizard-of-oz messaging proxy, implemented in Twilio Studio [31]

wizard to use their own phone number for sending messages. However, to maintain privacy and allow severability after the close of the study, we implemented an SMS-forwarding proxy with Twilio Studio [31], illustrated in Figure 1. This implementation required only a few dozen lines of JavaScript code to create the link between each participant’s phone number to their proxy phone number for appropriate routing of messages to and from the wizard’s phone number.

Because WOZ methods are primarily focused on research in a lab setting, we encountered a number of challenges adapting WOZ methods to our deployment study, which we describe below.

3.2.1 Challenge 1: Expectation of 24/7 availability. WOZ studies often require continuous effort and attention from the wizard while the user is interacting with the system, to ensure the appearance of a functioning prototype. With a deployment study, the system need to be at least somewhat functional at all times, which would put undue burden on the wizard due to a biological need for sleep and food, as well as other familial and academic obligations.

Because the envisioned interactions relied on one primary agent-initiated conversation per day, we sought to align the wizard’s schedules with when users were available and preferred to receive messages. During enrollment, we asked users to fill out a short form to determine their 1-hour long slot to receive messages. This also enabled one wizard to be assigned to multiple users at once, with the intention of minimizing the number of users to send responses to at once.

Importantly, to create a somewhat chatbot-like experience even when the wizard was not available, we configured Twilio to send a brief, automatic reply to let users know that t2.coach received their message and would respond within 24 hours. This type of automatic reply is uncommon when texting with a human conversational partner, and while not particularly informative, was meant to create the feeling of interacting with an automated system. Users often received automatic replies when asking questions outside of the designated times, or if the user was considerably delayed in replying to a message from t2.coach.

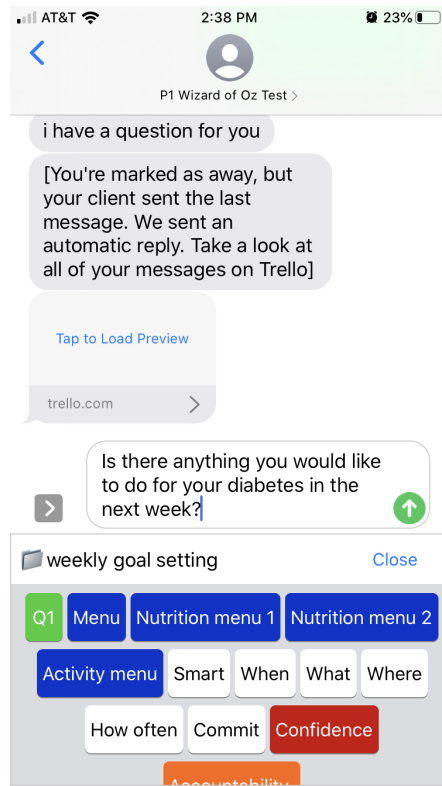


Figure 3: A screenshot of the WordBoard application, configured to allow the wizard to more easily send scripted messages

3.2.2 Challenge 2: Managing conversations with multiple users. While a WOz study in the lab would usually happen with the participant and wizard working synchronously, with a deployment study there is a need to enroll multiple participants in the study concurrently. To help the wizard manage their work across multiple participants, we utilized Trello [32], a project-management platform, which served as a central dashboard for the wizard to organize their work. The wizard used cards to schedule their needed outreach and outstanding items for each participant. In addition, we integrated Trello with the Twilio messaging set up. For example, we let the wizard update their availability on Trello in real time, which determined whether users would receive the automatic reply described in Challenge 1. Trello offers a straightforward API, and this integration was accomplished with a few lines of JavaScript.

3.2.3 Challenge 3: Imitating quick responses with slow-typing humans. When interacting with a chatbot, there is an expectation of relatively quick replies, as opposed to text messaging with a human conversational partner. Instead of having the wizard type out responses, we configured WordBoard, an existing iOS keyboard application with configurable text short-cuts [33]. As shown in Figure 3, the wizard could select an appropriate scripted response from the set of options to send to the participant.

4 RESULTS AND DISCUSSION

All 13 participants were able to complete the study without issues using the WOz text messaging setup (one participant dropped out for family reasons).

4.1 Addressing the Challenges of WOz in the Wild

To manage user’s expectations for the 24/7 availability of a chatbot, we scheduled a 1-hour slot of time with each participant and added automatic replies when the wizard was not available. In practice, however, we found that participants led busy and complex lives, and were not always available to respond to messages at a consistent time each day. As shown in Table 1, users responded sometime had a considerable delay in their responses — a median of 2.41 minutes and over 7 hours in some instances. This suggests that a simple strategy of assigning a single time slot at the same time each day may not be sufficient, and other scheduling approaches may be necessary to let participants adjust, cancel, or reschedule their time for daily messages. Still, when participants were available in their allotted time, the staggered schedule helped the wizard handle multiple participants at the same time, without needing to simultaneously send messages to more than two users at once.

The wizard, on the other hand, replied with a median delay of just under a minute. While still not as quick as a chatbot, these responses were considerably faster than users’ responses. With the help of Trello, text shortcuts and other automations, we found that a single wizard (working as a full-time PhD student) was able to work with up to 5 participants concurrently enrolled in the study. For a brief period of time when more than 5 participants were enrolled, response delays from the wizard increased noticeably. Future research could more precisely explore the workload necessary for WOz deployment studies, which likely relate to the complexity of the agent being tested and anticipated length of conversations.

4.2 Design Insights

Importantly, the WOz study surfaced a number of insights that may not have been uncovered with usability testing in a lab setting. Because we observed participants taking a considerable amount of time to respond to messages (Table 1) we made considerable changes to the BAP script as a part of this iterative design phase. While BAP was designed for human-to-human conversations in a single session, we found that the number of conversational steps was too long. Because of the delay, many participants were not able to finish the initial goal setting conversation on the first day, let alone the scheduled hour overlapping with the wizard. To incorporate this feedback, we substantially reduced to the length of scripted dialogs to focus on the core steps of goal setting. Importantly, this was an insight that would have been difficult to glean from a lab experiment, as users would have been more likely to complete the full conversation in a single setting under supervision of the researchers, without the distractions of daily life.

Second, the initial version of the script also included messages to remind users about the current active conversation if they were idle for more than an hour. Because users regularly took a long time to reply, these messages were unnecessary and also increased

Table 1: Average delay in responses to incoming text messages, in minutes

	Wizard	User
Mean response delay ($p < 0.01$)	4.45 minutes	35.44 minutes
Median	0.92 minutes	2.41 minutes
Range	0 to 234.53 minutes	0.02 to 428.30 minutes

annoyance among early participants (one replied “please stop”) and so we removed it from the script.

Third, while we expected participants to ask questions and initiate interactions with t2.coach, we found that users rarely initiated conversations or asked questions unprompted. Users initiated interaction with t2.coach only 16% of the time, and many of these were in the first session, when we asked each participant to send an initial question to t2.coach. It appeared that participants were willing to ask questions of t2.coach in the controlled setting during enrollment, but were less likely to follow through and ask additional questions in the wild. Based on this lack of utilization, we did not pursue adding additional question answering capabilities to the final implemented version of t2.coach.

4.3 Limitations and Future Work

The discussion in this case study comes from a study with a small sample of 13 individuals over a two-week deployment study. We recruited participants who were often less tech savvy than other user WOz and conversational agent studies, and our findings may not generalize to other populations.

While we are able to glean insights from a WOz deployment study, the approach is no doubt more resource-intensive than laboratory WOz studies. There are other design approaches that have been used to gain insights in a deployment setting, like technology probes [12]. Technology probes are simple, flexible applications with less functionality than a fully developed prototype, and are often used in early stages of design. Unlike longitudinal WOz, probes can be deployed and gather data for long periods of time without the need for direct researcher involvement. While probes can lead to useful design insights [11], they are applicable at different stages in the design process [3]. A probe is necessarily further removed from the functionality of a fully developed system than a WOz prototype, making WOz more useful for later in the design process. Given the resources required for longitudinal WOz studies, future work could more closely examine the costs and benefits for iterative design.

5 CONCLUSION

We argue for the need for wizard-of-oz (WOz) methods that can be deployed longitudinally in field studies, to complement existing approaches to the iterative design of conversational agents. We described our approach to addressing some of the key challenges posed in adapting WOz methods to a deployment setting, and described lessons learned from a deployment study that would have been unlikely to emerge in a lab setting. We hope that future researchers can build on these methods to offer improved approaches to WOz in the wild.

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